The Volatility Range Expansion Factor

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Text below is X's own machine translation from the Chinese original.

Tested another interesting quantitative factor called the Volatility Range Expansion Factor (below is the test chart)

The logic is still very intuitive: the market's true volatility isn't just about the simple high-low price difference of the day, but also needs to consider the implied volatility brought by "gaps"

Therefore, we incorporate three dimensions into consideration—the daily range (HIGH-LOW), the distance from yesterday's close to the high, and the distance from yesterday's close to the low, and take the maximum of them to depict the day's true volatility range

The factor formula is as follows:

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-MEAN(MAX(MAX((HIGH-LOW),ABS(DELAY(CLOSE,1)-HIGH)),ABS(DELAY(CLOSE,1)-LOW)),12)
```

Essentially, it's capturing the state transition of the market under implied volatility expansion. When volatility rapidly amplifies, it's usually accompanied by irrational trading and overreactions, which also provide potential trading opportunities for subsequent price corrections or trend continuations

The test is very simple, just two lines of code to run the full backtest, everyone can go test it out

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Attached chart 1 — "Cumulative Long Returns (Rebalance Period = 1, Log Scale = False)", x-axis running from 2005-01-21 to 2026. Three series plotted: Q-Top Cumulative Return, Q-3 Cumulative Return, Q-Bottom Cumulative Return. The top quantile ends near 80x, the middle near 20x and the bottom near 12x, with the ordering monotone throughout. The top quantile is broadly flat until a sharp step up around 2015, drifts through 2016–2023, then rises steeply into 2024–2026.

Attached chart 2 — "Cumulative Rank IC (Horizon Period = 1)", same date range. The cumulative Rank IC oscillates around zero from 2005 until roughly 2013, then trends upward — reaching about 9 by 2017, holding a plateau through 2020, dipping in 2021, and climbing again past 20 thereafter, on a y-axis scaled to above 40.

Original Chinese opening line, for reference: Volatility Range Expansion Factor